

The Local-Multiplier Idempotence Question

A negative answer in arXiv:math/0509437

Automated Mathematics Run fa_banach_001, lane 0

13 August 2026

Status: literature already answered (negative). No new theorem is claimed. Ara and Mathieu construct unital, primitive AF algebras whose second local multiplier algebra is strictly larger than the first, refuting the universal equality stated in Frank–Paulsen.

1 The source question

For a C^* -algebra A , let $M_{\text{loc}}(A)$ denote its local multiplier algebra. Frank and Paulsen [1] state on page 15 that one of the main open questions in the theory is whether

$$M_{\text{loc}}(A) = M_{\text{loc}}(M_{\text{loc}}(A)) \tag{1}$$

for every C^* -algebra A .

In the theory of local multiplier C^* -algebras the problem whether $M_{\text{loc}}(A)$ coincides with $M_{\text{loc}}(M_{\text{loc}}(A))$ for any C^* -algebra A is one of the main open questions, cf. [1, 27]. Set $M_{\text{loc}}^{k+1}(A) = M_{\text{loc}}(M_{\text{loc}}^k(A))$, which is called the $(k+1)$ -order local multiplier algebra of A . We show that any higher order local

Figure 1: Frank–Paulsen, arXiv:math/0106131, PDF page 15.

Their subsequent theorem places every iterated local multiplier algebra inside the same injective envelope, but it does not prove equality in (1).

2 The later counterexample

Ara and Mathieu [2] revisit the same equality. Their abstract says they construct an AF algebra for which the two local multiplier algebras do not agree, thereby answering the question raised by Pedersen. In the introduction they label (1) as Question (2), explicitly say that its answer is negative, and state:

Theorem 1.1 (Ara–Mathieu). There exist unital, primitive AF algebras A such that

$$M_{\text{loc}}(M_{\text{loc}}(A)) \neq M_{\text{loc}}(A).$$

In general, however, it turns out that the answer to Question (2) is negative. In this paper we provide a class of examples to this effect. Our main result is the following.

Theorem 1.1. *There exist unital, primitive AF-algebras A such that $M_{\text{loc}}(M_{\text{loc}}(A)) \neq M_{\text{loc}}(A)$.*

Figure 2: Ara–Mathieu, arXiv:math/0509437, PDF page 2.

This is a literal counterexample to the universal quantifier in (1). The answer is therefore complete and negative, already in the restrictive class of unital primitive AF algebras.

3 Provenance, scope, and verification

The local registry and attempt indexes contained no prior row for arXiv:math/0106131. A bounded exact-formula and keyword search located arXiv:math/0509437 and the later survey arXiv:1110.6858, which confirms that several such examples are now known. The displayed question and theorem were then checked directly in the two primary PDFs included in this packet.

The other source signals—classification of relatively injective C^* -algebras and characterization of algebras satisfying $M_{\text{loc}}(A) = I(A)$ —are broader programs and are not claimed to be resolved by the cited counterexample. This packet clears only the exact iterated-local- multiplier equality.

References

- [1] M. Frank and V. I. Paulsen, *Injective Envelopes of C^* -algebras as Operator Modules*, arXiv:math/0106131.
- [2] P. Ara and M. Mathieu, *A not so simple local multiplier algebra*, arXiv:math/0509437 (2005).